Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q02] | Part A (Q1-Q5)

For integers $0 < v < u$, let $A = (u, v)$. Reflect successively across $y = x$, the y-axis, the x-axis, and the y-axis to obtain B, C, D, E. If polygon $ABCDE$ has area 451, find $u + v$.

整数满足 $0 < v < u$，取 $A = (u, v)$。依次关于 $y = x$、y 轴、x 轴、y 轴反射，得到 B, C, D, E。若五边形面积为 451，求 $u + v$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2001-II-Q04] | Part A (Q1-Q5)

Points P, Q lie respectively on $8y = 15x$ and $10y = 3x$, with midpoint $(8, 6)$. If $PQ = m/n$ reduced, find $m + n$.

P, Q 分别位于 $8y = 15x$、$10y = 3x$，其中点为 $(8, 6)$。若 $PQ = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2009-I-Q04] | Part A (Q1-Q5)

In parallelogram $ABCD$, $M \in AB, N \in AD$ satisfy $AM/AB = 17/1000, AN/AD = 17/2009$. Let $P = AC \cap MN$. Find AC/AP.

平行四边形 $ABCD$ 中，$M \in AB, N \in AD$ 满足 $AM/AB = 17/1000, AN/AD = 17/2009$。令 $P = AC \cap MN$，求 AC/AP。

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[AIME-2009-II-Q03] | Part A (Q1-Q5)

Rectangle $ABCD$ has $AB = 100$ and midpoint E of AD. If $AC \perp BE$, find the greatest integer smaller than AD.

矩形 $ABCD$ 中 $AB = 100$，E 为 AD 中点。若 $AC \perp BE$，求小于 AD 的最大整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2011-I-Q02] | Part A (Q1-Q5)

Rectangle $ABCD$ has $AB = 12, BC = 10$. Interior points satisfy $BE = 9, DF = 8, BE \parallel DF, EF \parallel AB$, and line BE meets segment AD. Write $EF = m\sqrt{n} - p$ with squarefree n, and find $m + n + p$.

矩形 $ABCD$ 中 $AB = 12, BC = 10$。内部点 E, F 满足 $BE = 9, DF = 8, BE \parallel DF, EF \parallel AB$，且直线 BE 交线段 AD。若 $EF = m\sqrt{n} - p$ 且 n 无平方因子，求 $m + n + p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2011-I-Q03] | Part A (Q1-Q5)

Line L has slope $5/12$ through $A = (24, -1)$; line M is perpendicular to it through $B = (5, 6)$. Use L, M as new x, y axes, oriented positively toward A, B. Find the sum of the new coordinates of the old point $(-14, 27)$.

直线 L 过 $A = (24, -1)$、斜率为 $5/12$；直线 M 过 $B = (5, 6)$ 且垂直于 L。以两线为新横纵轴，正方向分别指向 A, B。求原坐标点 $(-14, 27)$ 的新坐标之和。

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[AIME-2011-II-Q02] | Part A (Q1-Q5)

In square $ABCD$, nonvertex points E on AD and F on BC satisfy $BE = EF = FD = 30$. Find the square area.

正方形 $ABCD$ 的边 AD, BC 上分别有非顶点 E, F，满足 $BE = EF = FD = 30$。求正方形面积。

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[AIME-2013-II-Q04] | Part A (Q1-Q5)

Construct equilateral triangle ABC from $A=(1,0), B=(2,2\sqrt{3})$, choosing C in the first quadrant. Its center (x,y) has $xy=p\sqrt{q}/r$, with coprime p,r and squarefree q. Find $p+q+r$.

以 $A=(1,0), B=(2,2\sqrt{3})$ 作等边三角形，取第一象限顶点 C。中心为 (x,y)，若 $xy=p\sqrt{q}/r$，p,r 互质、q 无平方因子，求三者和。

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[AIME-2015-I-Q04] | Part A (Q1-Q5)

Points A, B, C are collinear in that order with $AB = 16, BC = 4$. Equilateral triangles ABD, BCE lie on the same side of AC. Let M be the midpoint of AE and N the midpoint of CD. If triangle BMN has area x, find x^2.

A, B, C 依次共线，$AB = 16, BC = 4$。等边三角形 ABD, BCE 位于 AC 同侧。M, N 分别是 AE, CD 中点。若 BMN 面积为 x，求 x^2。

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[AIME-2018-II-Q04] | Part A (Q1-Q5)

A convex equiangular octagon has consecutive labels $CAROLINE$, with $CA = RO = LI = NE = \sqrt{2}$ and the other four sides one. Reconnect the same vertices in order $CORNELIA$, forming six bounded nonoverlapping triangles. If their total area is reduced a/b, find $a + b$.

凸等角八边形按 $CAROLINE$ 排列，$CA = RO = LI = NE = \sqrt{2}$，其余四边均为一。将原顶点改按 $CORNELIA$ 顺序连接，围成六个不重叠的有界三角形区域。若总面积为最简分数 a/b，求 $a + b$。

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[AIME-2019-II-Q01] | Part A (Q1-Q5)

On the same side of $AB = 9$, distinct points C, D satisfy $AC = BD = 17, BC = AD = 10$. The overlapping triangle regions ABC, BAD have area reduced m/n. Find $m + n$.

在线段 $AB = 9$ 同侧有不同点 C, D，满足 $AC = BD = 17, BC = AD = 10$。两三角形区域重叠面积为最简分数 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2020-II-Q04] | Part A (Q1-Q5)

A clockwise rotation through $m \in (0, 180)$ degrees about (x, y) maps $A = (0, 0), B = (0, 12), C = (16, 0)$ respectively to $(24, 18), (36, 18), (24, 2)$. Find $m + x + y$.

绕 (x, y) 顺时针旋转 m 度，$0 < m < 180$，将 $(0, 0), (0, 12), (16, 0)$ 依次变为 $(24, 18), (36, 18), (24, 2)$。求 $m + x + y$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

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[AIME-2021-I-Q02] | Part A (Q1-Q5)

Rectangle $ABCD$ has $AB = 3, BC = 11$. A second rectangle, in cyclic order $AECF$, shares opposite vertices A, C and has $AF = 7, FC = 9$, with F on the opposite side of line AC from B. Its overlap with $ABCD$ has area m/n reduced. Find $m + n$.

矩形 $ABCD$ 中 $AB = 3, BC = 11$。另一矩形按 A, E, C, F 顺序取顶点，共用对角顶点 A, C，且 $AF = 7, FC = 9$；F 与 B 位于直线 AC 两侧。两矩形重叠面积约分为 m/n，求 $m + n$。

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[AIME-2023-I-Q05] | Part A (Q1-Q5)

A square $ABCD$ is inscribed in a circle. A point P on that circle satisfies $PA\,PC = 56$ and $PB\,PD = 90$. Determine the square area.

正方形 $ABCD$ 内接于一圆，圆上点 P 满足 $PA\,PC = 56, PB\,PD = 90$。求正方形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2024-I-Q05] | Part A (Q1-Q5)

Rectangles $ABCD$ and $EFGH$ have consecutive side lengths $(107, 16)$ and $(184, 17)$. Along one line the points occur in order D, E, C, F; the rectangles lie on opposite sides of this line. If A, D, H, G are concyclic, determine CE.

矩形 $ABCD, EFGH$ 的连续两边长分别为 $(107, 16)$、$(184, 17)$。D, E, C, F 依次共线，两矩形位于该线两侧。若 A, D, H, G 共圆，求 CE。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2025-II-Q01] | Part A (Q1-Q5)

Collinear points A, B, C, D, E, F occur in that order, with $AC = 26, BD = 22, CE = 31, DF = 33, AF = 73$. An off-line point G has $CG = 40, DG = 30$. Find $[BGE]$.

共线点 A, B, C, D, E, F 依次排列，$AC = 26, BD = 22, CE = 31, DF = 33, AF = 73$。线外点 G 满足 $CG = 40, DG = 30$。求 BGE 面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2026-II-Q03] | Part A (Q1-Q5)

A simple nonconvex pentagon $ABCDE$ has angles $A = E = 90^\circ, B = D = 45^\circ$, with B, C, D on the same side of AE. Given $DE < AB$, $AE = 20$, $BC = 14\sqrt{2}$, integer $AB < 2026$, and area divisible by 16, count possible values of AB.

简单非凸五边形 $ABCDE$ 中，$A = E = 90^\circ, B = D = 45^\circ$，$B, C, D$ 位于 AE 同侧。已知 $DE < AB, AE = 20, BC = 14\sqrt{2}$，$AB$ 为小于 2026 的整数，面积为 16 的整数倍。求 AB 的可能值个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2000-II-Q08] | Part B (Q6-Q10)

In trapezoid $ABCD$, BC is perpendicular to the parallel bases AB, CD, and diagonals AC, BD are perpendicular. Given $AB = \sqrt{11}, AD = \sqrt{1001}$, determine BC^2.

梯形 $ABCD$ 中，BC 垂直两底 AB, CD，对角线 AC, BD 也互相垂直。已知 $AB = \sqrt{11}, AD = \sqrt{1001}$，求 BC^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2004-II-Q07] | Part B (Q6-Q10)

Fold rectangle $ABCD$ so B lands at $B' \in AD$. The crease meets AB, CD at E, F, with $AE = 8, EB = 17, CF = 3$. If the rectangle perimeter is m/n reduced, find $m + n$.

矩形 $ABCD$ 折叠后 B 与 AD 上点 B' 重合，折痕交 AB, CD 于 E, F。已知 $AE = 8, EB = 17, CF = 3$。若矩形周长约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2005-I-Q10] | Part B (Q6-Q10)

Triangle ABC has area 70, $B = (12, 19)$, $C = (23, 20)$, and $A = (p, q)$. Its median to BC lies on a line of slope -5. Maximize $p + q$.

三角形 ABC 面积为 70，$B = (12, 19), C = (23, 20), A = (p, q)$。到 BC 的中线所在直线斜率为 -5，求 $p + q$ 最大值。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2006-I-Q10] | Part B (Q6-Q10)

Eight radius-$1/2$ disks have centers $(i+1/2, j+1/2)$ for $i, j \in \{0,1,2\}$ except $(i,j) = (2,2)$. A slope-three line bisects their total area. Write it as $ax = by + c$ in relatively prime positive integers and find $a^2+b^2+c^2$.

八个半径 $1/2$ 的圆盘，圆心为 $(i+1/2, j+1/2)$，其中 $i, j = 0,1,2$，但去掉 $(2,2)$。一条斜率三的直线平分总面积，写成互质正整数方程 $ax = by + c$，求 $a^2+b^2+c^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2006-II-Q06] | Part B (Q6-Q10)

In unit square $ABCD$, points $E \in BC, F \in CD$ form equilateral triangle AEF. A smaller axis-parallel square inside $ABCD$ has vertex B and its opposite vertex on AE. Its side is $(a - \sqrt{b})/c$ in squarefree form. Find $a + b + c$.

单位正方形 $ABCD$ 中，$E \in BC, F \in CD$ 使 AEF 为等边三角形。内部另一边平行的小正方形以 B 为一顶点、对顶点在 AE 上，边长为 $(a - \sqrt{b})/c$，求 $a + b + c$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2009-II-Q10] | Part B (Q6-Q10)

Points A, B, C have $AB = 5, BC = 12, AC = 13$. A fourth point D satisfies $\angle BAD = \angle CAD$ and $\angle ACB = \angle DCB$, with nondegenerate rays. If $AD = p\sqrt{r}/q$ is reduced and r squarefree, find $p + q + r$.

$AB = 5, BC = 12, AC = 13$，另有点 D 满足 $\angle BAD = \angle CAD, \angle ACB = \angle DCB$，各射线不退化。若 $AD = p\sqrt{r}/q$ 为约简形式，求 $p + q + r$。

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[AIME-2010-II-Q09] | Part B (Q6-Q10)

In regular hexagon $ABCDEF$, let G, H, I, J, K, L be the midpoints of AB, BC, CD, DE, EF, FA. Segments AH, BI, CJ, DK, EL, FG enclose a smaller regular hexagon. Its area divided by the original area is m/n in lowest terms. Find $m+n$.

正六边形 $ABCDEF$ 各边 AB, BC, CD, DE, EF, FA 的中点依次为 G, H, I, J, K, L。线段 AH, BI, CJ, DK, EL, FG 围成一个较小正六边形，面积比为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2013-I-Q07] | Part B (Q6-Q10)

A rectangular box measures 12 by 16 by m/n, with reduced positive m/n. The centers of three faces meeting at one corner form a triangle of area thirty. Find $m+n$.

长方体的三尺寸为十二、十六及最简正分数 m/n。同一顶点处三个面的中心构成面积三十的三角形，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2014-I-Q10] | Part B (Q6-Q10)

A unit disk rolls without slipping outside a fixed radius-five disk, initially tangent at marked point A. Let E be the fixed center. Stop after the rolling disk rotates once relative to the ground, and let B be the new position of its marked point. Reduce $\sin^2 \angle BEA = m/n$ and find $m+n$.

单位圆盘沿固定半径五的圆盘外侧无滑动滚动，初始切点标为 A，固定圆心为 E。当活动圆盘相对地面自转一整圈时停止，标记点到达 B。将 $\sin^2 \angle BEA$ 约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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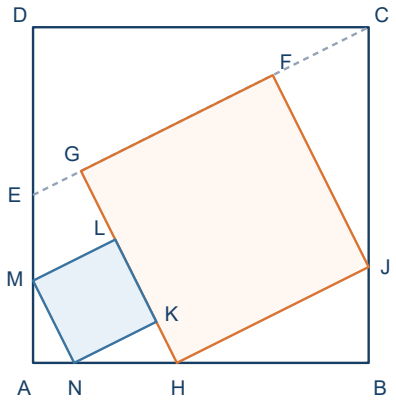
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[AIME-2015-I-Q07] | Part B (Q6-Q10)

In square $ABCD$, E is the midpoint of AD. Square $FGHJ$ has F, G on CE, H on AB, and J on BC. Square $KLMN$ has K, L on GH, M on AD, and N on AB; vertex names for each square follow its boundary. Given $[KLMN] = 99$, find $[FGHJ]$.

正方形 $ABCD$ 中，E 是 AD 中点。正方形 $FGHJ$ 的 F, G 在 CE 上、H 在 AB 上、J 在 BC 上。正方形 $KLMN$ 的 K, L 在 GH 上、M 在 AD 上、N 在 AB 上；各正方形顶点均按边界顺序命名。已知 $KLMN$ 面积为 99，求 $FGHJ$ 面积。



Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2016-II-Q07] | Part B (Q6-Q10)

Two concentric, side-parallel squares have areas 2016 and a smaller positive integer. A third square has one vertex on each side of the outer square, while each vertex of the inner square lies on a side of the third. Find the difference between the largest and smallest integral areas of the third square.

两个同心且对应边平行的正方形面积分别为 2016 和某个较小正整数。第三个正方形的四个顶点分别在外正方形四边上，内正方形每个顶点又在第三个正方形的一边上。求第三正方形面积的最大、最小整数值之差。

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[AIME-2017-II-Q10] | Part B (Q6-Q10)

Rectangle $ABCD$ has $AB = 84, AD = 42$. Let M be the midpoint of AD, N the point on AB with $AN = 28$, and $O = CM \cap DN$. A segment BP, with P on the boundary of quadrilateral $BCON$, splits its area equally. Find $[CDP]$.

矩形 $ABCD$ 中 $AB = 84, AD = 42$。M 为 AD 中点，N 在 AB 上且 $AN = 28$，$O = CM \cap DN$。线段 BP 将四边形 $BCON$ 面积平分，P 在其边界上。求 $[CDP]$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2018-I-Q08] | Part B (Q6-Q10)

An equiangular convex hexagon $ABCDEF$ has $AB = 6, BC = 8, CD = 10, DE = 12$. If d is the largest diameter of a disk contained in it, find d^2.

凸等角六边形 $ABCDEF$ 满足 $AB = 6, BC = 8, CD = 10, DE = 12$。设可完全放入其中的圆盘最大直径为 d，求 d^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2018-II-Q09] | Part B (Q6-Q10)

From $[0,27]\times[0,23]$, remove each corner right triangle with horizontal leg eight and vertical leg six, obtaining octagon $ABCDEFGH$ starting at $A=(0,6)$ and $B=(8,0)$ around the boundary. Let J be the midpoint of AH. Fan-triangulate from J into seven triangles using the other six vertices B through G. Find the area of their centroid polygon.

从矩形 $[0,27]\times[0,23]$ 各角去掉水平直角边八、竖直直角边六的三角形，得到依次以 $A=(0,6), B=(8,0)$ 开始的八边形 $ABCDEFGH$。J 为 AH 中点，从 J 连至 B 至 G 将区域分为七个三角形，求各重心组成多边形的面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2021-I-Q09] | Part B (Q6-Q10)

In an isosceles trapezoid $ABCD$ with parallel bases $AB < CD$ and legs $AD = BC$, the distances from A to the lines BC, CD, BD are $15, 18, 10$. If its area is K, determine $\sqrt{2}K$.

等腰梯形 $ABCD$ 的底边 $AB \parallel CD$、$AB < CD$，腰 $AD = BC$。点 A 到直线 BC, CD, BD 的距离依次为 $15, 18, 10$。若面积为 K，求 $\sqrt{2}K$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

Source / 原题来源

Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q08] | Part B (Q6-Q10)

A rhombus $ABCD$ has acute angle BAD. Point P lies on its incircle and is at distances $9, 5, 16$ from the lines DA, AB, BC, respectively. Find the rhombus perimeter.

菱形 $ABCD$ 的 $\angle BAD$ 为锐角。内切圆上点 P 到直线 DA, AB, BC 的距离依次为 $9, 5, 16$。求菱形周长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-I-Q08] | Part B (Q6-Q10)

For real k, the system $|25+20i-z|=5$, $|z-4-k|=|z-3i-k|$ has exactly one complex solution. The sum of all such k is m/n in lowest terms. Find $m+n$.

实数 k 使复数方程组 $|25+20i-z|=5, |z-4-k|=|z-3i-k|$ 恰有一解。所有这样的 k 之和为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-I-Q10] | Part B (Q6-Q10)

Triangle ABC has $AB = 13, BC = 14, CA = 15$. Rotate it about its circumcenter to $A'B'C'$ so $A'C' \perp BC$ and A', B lie on opposite sides of $B'C'$. Find the nearest integer to area $AA'CC'BB'$.

三角形 ABC 边长为 $AB = 13, BC = 14, CA = 15$，绕外心旋转得 $A'B'C'$，使 $A'C' \perp BC$，且 A', B 位于 $B'C'$ 两侧。求六边形 $AA'CC'BB'$ 面积最接近的整数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q13] | Part C (Q11-Q15)

A vehicle starts at the crossing of perpendicular straight roads. It travels at speed 50 miles/hour on roads and 14 miles/hour elsewhere in the plane. The area reachable within six minutes is m/n square miles in lowest terms. Find $m+n$.

车辆从两条垂直直路交点出发，公路速度为每小时 50 英里，平面其他地方速度为每小时 14 英里。六分钟内可达区域面积约分为 m/n 平方英里，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2000-II-Q11] | Part C (Q11-Q15)

An isosceles trapezoid has lattice vertices, $A = (20, 100), D = (21, 107)$, with AB, CD its only parallel sides and no side horizontal or vertical. Sum the absolute values of every possible distinct slope of AB, write the result m/n reduced, and find $m + n$.

等腰梯形顶点均为整点，$A = (20, 100), D = (21, 107)$；仅 AB, CD 平行，所有边均不水平也不竖直。将 AB 所有不同可能斜率的绝对值相加，约分为 m/n，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2002-I-Q11] | Part C (Q11-Q15)

Inside the cube $[0,12]^3$, a light ray starts at $(0,0,0)$ in direction $(12,7,5)$ and reflects specularly at every face. At its first subsequent arrival at a cube vertex, its traveled length is $m\sqrt{n}$ with squarefree n. Find $m+n$.

在立方体 $[0,12]^3$ 内，光线从 $(0,0,0)$ 沿 $(12,7,5)$ 方向出发，在每个面镜面反射。首次再次到达立方体顶点时，路程为 $m\sqrt{n}$，其中 n 无平方因子，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2003-I-Q15] | Part C (Q11-Q15)

In ABC, $AB = 360, BC = 507, CA = 780$. Let M be the midpoint of CA, and let BD be the internal bisector of B meeting CA. Through D, the perpendicular to BD meets BC at F and BM at E. If $DE/EF = m/n$ reduced, find $m + n$.

三角形 ABC 边长为 $AB = 360, BC = 507, CA = 780$。$M$ 为 CA 中点，B 的内角平分线交 CA 于 D。过 D 作 BD 垂线交 BC 于 F、交 BM 于 E。若 $DE/EF = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2003-II-Q11] | Part C (Q11-Q15)

In right triangle ABC, $AC = 7, BC = 24$ and C is right. Let M be the midpoint of AB. Choose D on the same side of AB as C with $AD = BD = 15$. Write $[CDM] = m\sqrt{n}/p$ reduced with squarefree n, and find $m + n + p$.

直角三角形 ABC 中 $AC = 7, BC = 24$，直角在 C。M 为斜边 AB 中点，D 与 C 位于 AB 同侧且 $AD = BD = 15$。将 $[CDM]$ 写为最简 $m\sqrt{n}/p$，n 无平方因子，求参数和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2003-II-Q14] | Part C (Q11-Q15)

A convex equilateral hexagon $ABCDEF$ has $A=(0,0), B=(b,2)$, $\angle FAB = 120^\circ$, and opposite sides parallel: $AB \parallel DE, BC \parallel EF, CD \parallel FA$. Its six y-coordinates are $0, 2, 4, 6, 8, 10$ in some order. If area $= m\sqrt{n}$ with squarefree n, find $m+n$.

凸等边六边形 $ABCDEF$ 满足 $A=(0,0), B=(b,2), \angle FAB = 120^\circ$，且 $AB \parallel DE, BC \parallel EF, CD \parallel FA$。六个纵坐标恰为 $0, 2, 4, 6, 8, 10$。面积写为 $m\sqrt{n}$，n 无平方因子，求其和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2004-II-Q15] | Part C (Q11-Q15)

A 1024×1 strip is divided into unit squares numbered from the left. Ten times, fold its right half over and on top of its left half. In the final stack, how many squares are below the square originally numbered 942?

长一千零二十四、宽一的纸条分为单位方格，从左编号。连续十次将右半折到左半上方。最终堆叠中，原第九百四十二格下方有多少格?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2005-I-Q14] | Part C (Q11-Q15)

Points $A=(0,12), B=(10,9), C=(8,0), D=(-4,7)$ lie on different sides of one square. If its area is K, find $10K \bmod 1000$.

点 $A=(0,12), B=(10,9), C=(8,0), D=(-4,7)$ 分别位于同一正方形的不同边上。若其面积为 K，求 $10K$ 除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Working / 解题过程

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-II-Q12] | Part C (Q11-Q15)

In a side-900 square with center O, points A, E, F, B occur in order along one side, $EF = 400$, $AE < BF$, and $\angle EOF = 45^\circ$. If $BF = p + q\sqrt{r}$ in squarefree form, find $p + q + r$.

边长九百的正方形中心为 O，某边上依次为 A, E, F, B，$EF = 400, AE < BF, \angle EOF = 45^\circ$。若 $BF = p + q\sqrt{r}$，r 无平方素因子，求 $p + q + r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2006-I-Q14] | Part C (Q11-Q15)

A rigid tripod has three length-five legs with equal mutual angles and initially stands with its top four feet high. One leg loses its bottom foot while the mutual angles remain fixed. The new height is $h = m/\sqrt{n}$ with squarefree n. Find $\lfloor m + \sqrt{n} \rfloor$.

刚性三脚架三腿均长五英尺，两两夹角相同，原顶高四。某一腿底部断去一英尺而夹角保持不变。新高度为 $m/\sqrt{n}$，n 无平方素因子，求 $\lfloor m + \sqrt{n} \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q14] | Part C (Q11-Q15)

In regular octagon $A_1 \cdots A_8$, let M_i be the midpoint of A_iA_{i+1} for odd i. From these four points draw inward rays R_i with each consecutive pair perpendicular. Let $B_i = R_i \cap R_{i+2}$, using indices modulo eight. If $B_1B_3 = A_1A_2$, write $\cos(2\angle A_3M_3B_1) = m - \sqrt{n}$ with positive integers m, n, and find $m + n$.

正八边形 $A_1 \cdots A_8$ 中，奇数 i 对应的 M_i 为边 A_iA_{i+1} 中点。从四点向内作射线 R_i，相邻射线互相垂直，记 $B_i = R_i \cap R_{i+2}$，下标模八。若 $B_1B_3 = A_1A_2$，将 $\cos(2\angle A_3M_3B_1)$ 写成 $m - \sqrt{n}$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2011-II-Q13] | Part C (Q11-Q15)

In a side-12 square $ABCD$, point P is on AC with $AP > CP$. The circumcenters O_1, O_2 of ABP, CDP satisfy $\angle O_1PO_2 = 120^\circ$. If $AP = \sqrt{a} + \sqrt{b}$, find $a + b$.

边长十二的正方形 $ABCD$ 中，P 在对角线 AC 上且 $AP > CP$。ABP, CDP 的外心分别为 O_1, O_2，满足 $\angle O_1PO_2 = 120^\circ$。若 $AP = \sqrt{a} + \sqrt{b}$，求 $a + b$。

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坐标、向量与变换

[AIME-2012-I-Q13] | Part C (Q11-Q15)

An equilateral triangle has one vertex on each of three concentric circles of radii $3, 4, 5$. Its greatest area is $a + (b/c)\sqrt{d}$, with coprime positive b, c and squarefree d. Find $a + b + c + d$.

一个等边三角形的三个顶点分别在半径为三、四、五的同心圆上。其最大面积为 $a + (b/c)\sqrt{d}$，正整数 b, c 互质且 d 无平方因子，求四者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2012-II-Q13] | Part C (Q11-Q15)

Equilateral ABC has side $\sqrt{111}$. Points D_1, D_2 both satisfy $AD_i = \sqrt{111}, BD_i = \sqrt{11}$. On each base AD_i construct both equilateral triangles, giving four apexes $E_1, \ldots, E_4$. Find $\sum_{k=1}^{4} CE_k^2$.

等边三角形 ABC 边长为 $\sqrt{111}$。两点 D_1, D_2 均满足 $AD_i = \sqrt{111}, BD_i = \sqrt{11}$。在每条 AD_i 两侧分别作等边三角形，四顶点为 E_1 至 E_4。求 $\sum CE_k^2$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2013-I-Q12] | Part C (Q11-Q15)

Triangle PQR has angles $P = 75^\circ, Q = 60^\circ$. A unit regular hexagon $ABCDEF$ lies inside it, with AB on PQ, CD on QR, and another vertex on RP. Express the triangle area as $(a + b\sqrt{c})/d$, with coprime a, d and squarefree c, and find $a + b + c + d$.

三角形 PQR 的角为 $75^\circ, 60^\circ$。单位边长正六边形 $ABCDEF$ 位于内部，AB 在 PQ 上，CD 在 QR 上，另有一顶点在 RP 上。若三角形面积为 $(a + b\sqrt{c})/d$，a, d 互质、c 无平方因子，求四者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2013-II-Q13] | Part C (Q11-Q15)

An isosceles triangle has $AC = BC$. Point D on BC obeys $CD = 3BD$, and E is the midpoint of AD. Given $CE = \sqrt{7}, BE = 3$, write the triangle area $m\sqrt{n}$ with squarefree n and find $m + n$.

等腰三角形满足 $AC = BC$，边 BC 上点 D 满足 $CD = 3BD$，E 为 AD 中点。已知 $CE = \sqrt{7}, BE = 3$，若面积为 $m\sqrt{n}$ 且 n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2014-I-Q13] | Part C (Q11-Q15)

On the respective sides AB, BC, CD, DA of a square lie E, F, G, H. The segments EG, FH are perpendicular, both of length 34, and meet at P. The corner-region areas $AEPH, BFPE, CGPF, DHPG$ are in ratio $269 : 275 : 405 : 411$. Find the square area.

正方形四边 AB, BC, CD, DA 上分别有 E, F, G, H，EG, FH 垂直且均长三十四，交于 P。四角区域 $AEPH, BFPE, CGPF, DHPG$ 面积比为 $269 : 275 : 405 : 411$。求正方形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2014-II-Q11] | Part C (Q11-Q15)

Triangle RED has $RD = 1, \angle R = 75^\circ, \angle E = 45^\circ$. Let M be the midpoint of RD and take C on ED so $RC \perp EM$. Extend DE past E to A with $CA = AR$. If $AE = (a - \sqrt{b})/c$ with coprime a, c, find $a + b + c$.

三角形 RED 中 $RD = 1, \angle R = 75^\circ, \angle E = 45^\circ$，$M$ 为 RD 中点。在 ED 上取 C 使 $RC \perp EM$。延长 DE 过 E 至 A，使 $CA = AR$。若 $AE = (a - \sqrt{b})/c$、a, c 互质，求三者和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2015-II-Q11] | Part C (Q11-Q15)

Acute triangle ABC has circumcenter O. The line through O perpendicular to OB meets lines AB, BC at P, Q. Given $AB = 5, BC = 4, BQ = 9/2$, express $BP = m/n$ in lowest terms and find $m + n$.

锐角三角形 ABC 的外心为 O。过 O 作垂直 OB 的直线，交直线 AB, BC 于 P, Q。已知 $AB = 5, BC = 4, BQ = 9/2$。将 $BP = m/n$ 约为最简分数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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坐标、向量与变换

[AIME-2016-I-Q14] | Part C (Q11-Q15)

At each lattice point, center a circle of radius $1/10$ and an axis-aligned closed square of side $1/5$. The segment from $(0,0)$ to $(1001,429)$ intersects m squares and n circle circumferences, with tangency included. Find $m+n$.

每个整点为圆心作半径 $1/10$ 的圆周，并以该点为中心作边长 $1/5$、边平行坐标轴的闭正方形。从 $(0,0)$ 到 $(1001,429)$ 的线段与 m 个正方形、n 个圆周相交，切触也计入。求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2017-I-Q15] | Part C (Q11-Q15)

A right triangle has legs $5, 2\sqrt{3}$ and hypotenuse $\sqrt{37}$. Among equilateral triangles with one vertex on each side segment, write the smallest area as $m\sqrt{p}/n$, where m, n are coprime positive integers and p is squarefree. Find $m+n+p$.

直角三角形的两直角边为 $5, 2\sqrt{3}$，斜边为 $\sqrt{37}$。在每条边线段上各取一顶点构成等边三角形，将最小面积写为 $m\sqrt{p}/n$，其中 m, n 为互质正整数，p 无平方因子。求 $m+n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2017-II-Q15] | Part C (Q11-Q15)

Tetrahedron $ABCD$ has opposite edge pairs $AD = BC = 28$, $AC = BD = 44$, $AB = CD = 52$. Minimize $AX + BX + CX + DX$ over all spatial points X. If the minimum is $m\sqrt{n}$ with squarefree positive integer n, find $m + n$.

四面体 $ABCD$ 的对棱分别满足 $AD = BC = 28, AC = BD = 44, AB = CD = 52$。在空间任意点 X 中最小化 $AX + BX + CX + DX$。若最小值为 $m\sqrt{n}$，n 为无平方因子的正整数，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2019-II-Q11] | Part C (Q11-Q15)

Triangle ABC has $AB = 7, BC = 8, CA = 9$. One circle passes through B and is tangent to line AC at A; another passes through C and is tangent to line AB at A. Their second intersection is K. Write $AK = m/n$ reduced and find $m + n$.

三角形 ABC 中 $AB = 7, BC = 8, CA = 9$。一圆过 B 并在 A 与直线 AC 相切，另一圆过 C 并在 A 与 AB 相切。再交点为 K，若 $AK = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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[AIME-2019-II-Q13] | Part C (Q11-Q15)

A regular octagon is inscribed in a circle of area one. For an interior point P, the curvilinear regions formed by P, A_1, A_2 and their minor arc, and by P, A_3, A_4 and their minor arc, have areas $1/7, 1/9$. The analogous P, A_6, A_7 region has area $1/8 - \sqrt{2}/n$. Find positive integer n.

正八边形内接于面积一的圆，圆内点 P 与 A_1, A_2 及其劣弧围成区域面积 $1/7$，与 A_3, A_4 对应区域面积 $1/9$。P, A_6, A_7 对应面积为 $1/8 - \sqrt{2}/n$，求正整数 n。

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[AIME-2020-I-Q15] | Part C (Q11-Q15)

An acute triangle ABC has circumcircle ω and orthocenter H. The tangent at H to the circle through H, B, C cuts ω at X, Y. Suppose $HA = 3, HX = 2, HY = 6$. If $[ABC] = m\sqrt{n}$ with n squarefree, find $m + n$.

锐角三角形 ABC 的外接圆为 ω，垂心为 H。过 H, B, C 的圆在 H 处的切线交 ω 于 X, Y。已知 $HA = 3, HX = 2, HY = 6$。设面积为 $m\sqrt{n}$，n 无平方因子，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-II-Q12] | Part C (Q11-Q15)

An m-row, n-column grid, with odd $m, n > 1$, is numbered consecutively left to right in each row, starting at the top. Label 200 lies in row 1, and label 2000 in row m. Count ordered pairs (m, n) for which the line through those two cell centers crosses the interior of cell 1099.

一个 m 行 n 列单位方格阵中，$m, n > 1$ 均为奇数，从顶行起逐行由左向右连续编号。200 号在顶行，2000 号在底行。求使这两格中心连线穿过 1099 号格内部的有序对 (m, n) 数量。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-II-Q12] | Part C (Q11-Q15)

A convex quadrilateral has consecutive side lengths $5, 6, 9, 7$ and area 30. Its diagonals meet at acute angle θ. Write $\tan\theta = m/n$ reduced and find $m+n$.

凸四边形连续边长为 $5, 6, 9, 7$，面积为 30。两对角线的锐夹角为 θ。将 $\tan\theta$ 约分为 m/n，求 $m+n$。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q12] | Part C (Q11-Q15)

An equilateral triangle ABC has side 55. On BC, CA, AB lie D, E, F, with $BD = 7, CE = 30, AF = 40$. An interior point P satisfies $\angle AEP = \angle BFP = \angle CDP = \theta$. Find $\tan^2\theta$.

边长为 55 的等边三角形 ABC 中，D, E, F 分别在 BC, CA, AB 上，且 $BD = 7, CE = 30, AF = 40$。内点 P 满足 $\angle AEP = \angle BFP = \angle CDP = \theta$。求 $\tan^2\theta$。

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[AIME-2023-II-Q12] | Part C (Q11-Q15)

Triangle ABC has sides $AB = 13, BC = 14, CA = 15$. Let M be the midpoint of BC, and let line AM meet the circumcircle again at P, beyond M. The unique Q on AM satisfying $\angle PBQ = \angle PCQ$ has $AQ = m/\sqrt{n}$, with coprime positive integers m, n. Find $m + n$.

三角形 ABC 的边长为 $AB = 13, BC = 14, CA = 15$。$M$ 为 BC 中点，直线 AM 在 M 外侧再交外接圆于 P。线段 AM 上唯一满足 $\angle PBQ = \angle PCQ$ 的点为 Q。若 $AQ = m/\sqrt{n}$，正整数 m, n 互质，求 $m + n$。

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[AIME-2025-I-Q12] | Part C (Q11-Q15)

In the plane $x+y+z=75$, the inequalities $x-yz<y-zx<z-xy$ define three convex regions, only one bounded. If its area is $a\sqrt{b}$ with squarefree b, find $a+b$.

平面 $x+y+z=75$ 内，满足 $x-yz<y-zx<z-xy$ 的点组成三个凸区域，恰有一个有界。其面积为 $a\sqrt{b}$，b 无平方素因子，求 $a+b$。

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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-I-Q14] | Part C (Q11-Q15)

An equiangular pentagon has sum of squared side lengths 308 and sum of squared diagonals 800. Its squared perimeter is $m\sqrt{n}$, with squarefree n. Find $m+n$.

等角五边形的边长平方和为 308，对角线长平方和为 800。周长平方为 $m\sqrt{n}$，其中 n 无平方素因子，求 $m+n$。

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